Smart Audio Diagnostics - Analysis Report

21 Oct 2026, 21:52

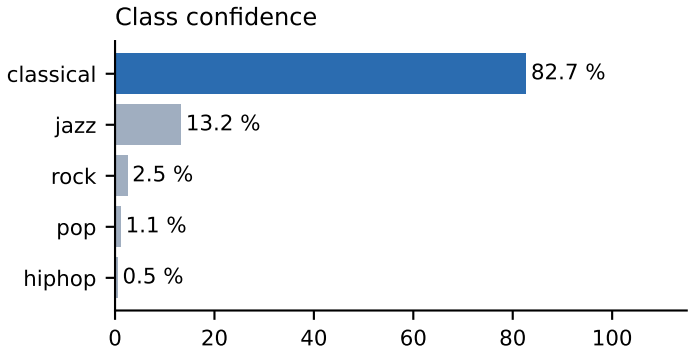
Group 09 · Digital Signal Processing · ECE, MNNIT Allahabad

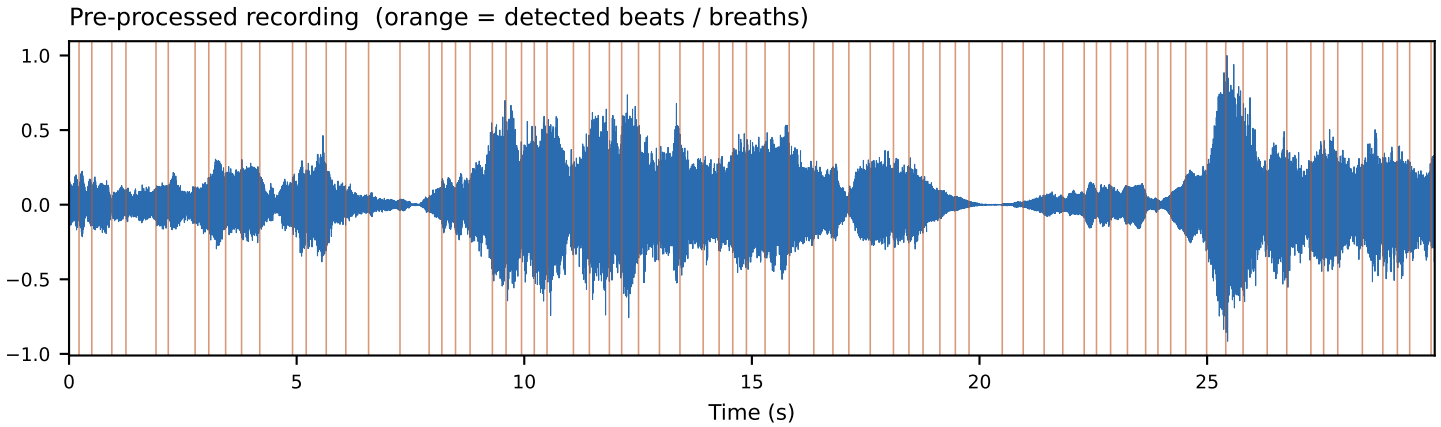
File	music_classical_01.wav
Signal type	Music - genre
Auto-detected as	music (85 % confidence)
Classifier	Random Forest (held-out accuracy 82.2 %)
Trained on	real:C:\dsp\data\gtzan\genres

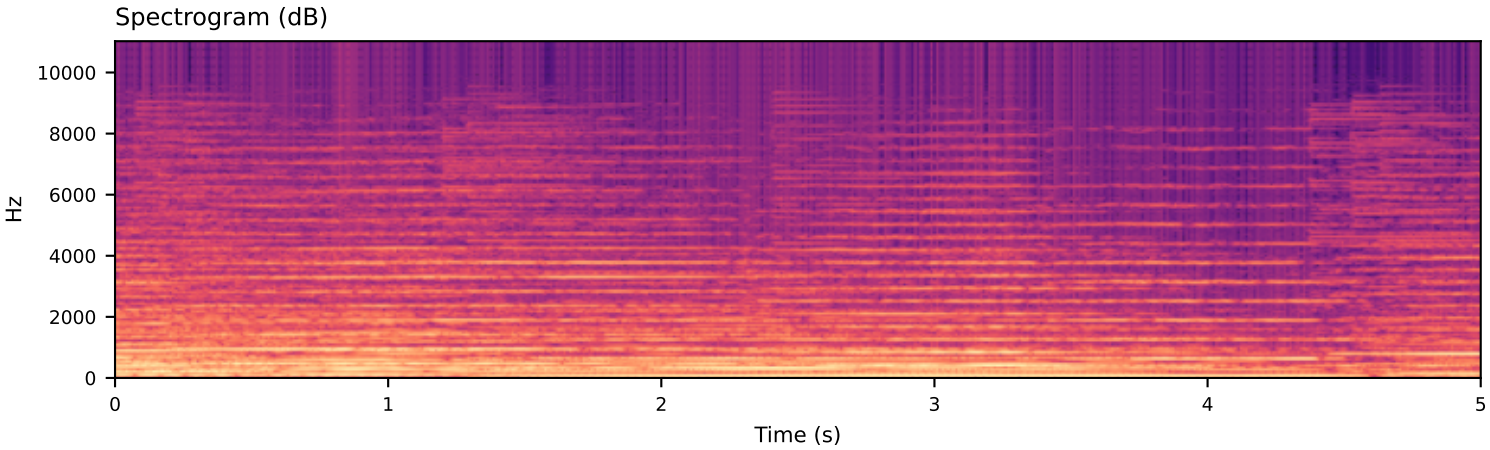
RESULT

CLASSICAL

Tempo
133 bpm







DSP feature	Value	DSP feature	Value
Sampling frequency	22050 Hz	Spectral roll-off 85%	628.4 Hz
Band-pass	30-10000 Hz	Spectral flatness	0.0008
Frame / hop	25 / 10 ms	Zero-crossing rate	0.0512
Mel filters	40	Short-time energy	0.01818
Spectral centroid	396.2 Hz	RMS	0.1289
Spectral bandwidth	464.9 Hz	MFCC-1 (mean)	-85.148